Exercise 2C

Numericals :-

2) $\boxed{S = ut + \frac{1}{2}at^2}$

$S = (10 \times 5) + \frac{1}{2} \times (5 \times (5)^2)$

$\Rightarrow 50 + \frac{1}{2} \times (5 \times 25)$

$\Rightarrow 50 + \frac{1}{2} \times 125$

$\Rightarrow 50 + 62.5$

$\Rightarrow 112.5\,m$

4)

(i) $v = u + at$

$v = 0 + 2 \times 5$

$= 10\,m/s$

(ii) $\boxed{S = ut + \frac{1}{2}at^2}$

$S = 0 \times 5 + \frac{1}{2} \times (2 \times (5)^2)$

$\Rightarrow \frac{1}{2} \times 50$

$\Rightarrow 25\,m$

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5) $u = 20 m/s$
$s = 10 cm = 0.1 m$

$\cancel{v^2 = u^2 + 2as}$
$\cancel{0 = 400 + 2as}$
$\cancel{\Rightarrow 400 + 2a \times 0.1}$

$$\boxed{v^2 = u^2 + 2as}$$

$v^2 = (20)^2 + 2a \times 0.1$
$0 = 400 + 0.2a$
$-400 = 0.2a$
$-4000 = 2a$
$a = \frac{-4000}{2}$

$= -2000 m/s^2$

$\therefore$ Retardation $= 2000 m/s^2$

8)
$u = 90 km/h = \frac{90000}{3600} \frac{m}{s} = 25 m/s$

$v = 0 m/s$
$a = -0.5 m/s^2$

(i) $\boxed{v = u + at}$

$\cancel{v = 0 \; 90 + -5 \; 20}$
$\cancel{= 85 m/s^2}$
$v = 25 - 5$
$= 20 m/s$

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(iii) $\boxed{v = u + at}$

$0 = 25 + -0.5t$

$-25 = -0.5t$

$t = \frac{-25}{-0.5}$

$= 50\,s$

10) $a = -25\,cm/s^2 = 0.25\,m/s$

$t = 20\,s$

(i) $\boxed{v = u + at}$

$0 = u - (0.25 \times 20)$

$0 = u - 5$

$u = 5\,m/s$

(ii)

$\boxed{s = ut + \frac{1}{2}at^2}$

$s = (5 \times 20) + \frac{1}{2}(-0.25 \times (20)^2)$

$\Rightarrow s = 100 + \frac{1}{2}\{-0.25 \times (20)^2\}$

$s = 100 + \frac{1}{2}\{-0.25 \times 400\}$

$s = 100 + \frac{1}{2} \times 100$

$= 150\,m$

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(ii) $\boxed{s = ut + \frac{1}{2}at^2}$

$270 = 0 + \frac{1}{2} \times a \times (3)^2$

$270 = \frac{1}{2} \times a \times 9$

$270 = \frac{9a}{2}$

$540 = 9a$

$\boxed{a = 60 m/s^2}$

$\boxed{v = u + at}$

$v = 0 + 60 \times 10$

$v = 600 m/s$

15) $u = 0$

$a = 2 m/s^2$

$t = 10 s$

$v = ?$

(i) $v = u + at$

$= 0 + 2 \times 10$

$= 20 m/s$

(ii) $v = u + at$

$0 = 20 + a \times 50$

$-20 = 50a$

$a = \frac{-20}{50}$

$= -0.4 m/s^2$

∴ Retardation $= 0.4 m/s^2$

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(iii)

SOL=

total distance $= \left(\frac{1}{2} \times 2 \times (10)^2\right) + \left(2000 \times 10\right) +$

$\left(\frac{1}{2} \times -0.4 \times 50\right)$

$= 100 + 20000 + 500\,m$

$= 20600\,m$

(iv) avg velocity $= \frac{\cancel{20600}^{23}}{\cancel{2}600_{13}} = 17.69\,m/s$

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